

Software Requirements Specification (SRS)

Multi-Format File Processing System

Document Control

Item	Details
Document Title	SRS: Multi-Format File Ingestion & Document Processing System
Supported File Formats	PPT/PPTX, Excel (XLSX/XLS/CSV), PDF, Images (PNG/JPG), EML Emails
Primary AI Inference Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Systems Engineers, Integration Architects, QA Testers

1. Functional Scope & Individual Flow Diagram

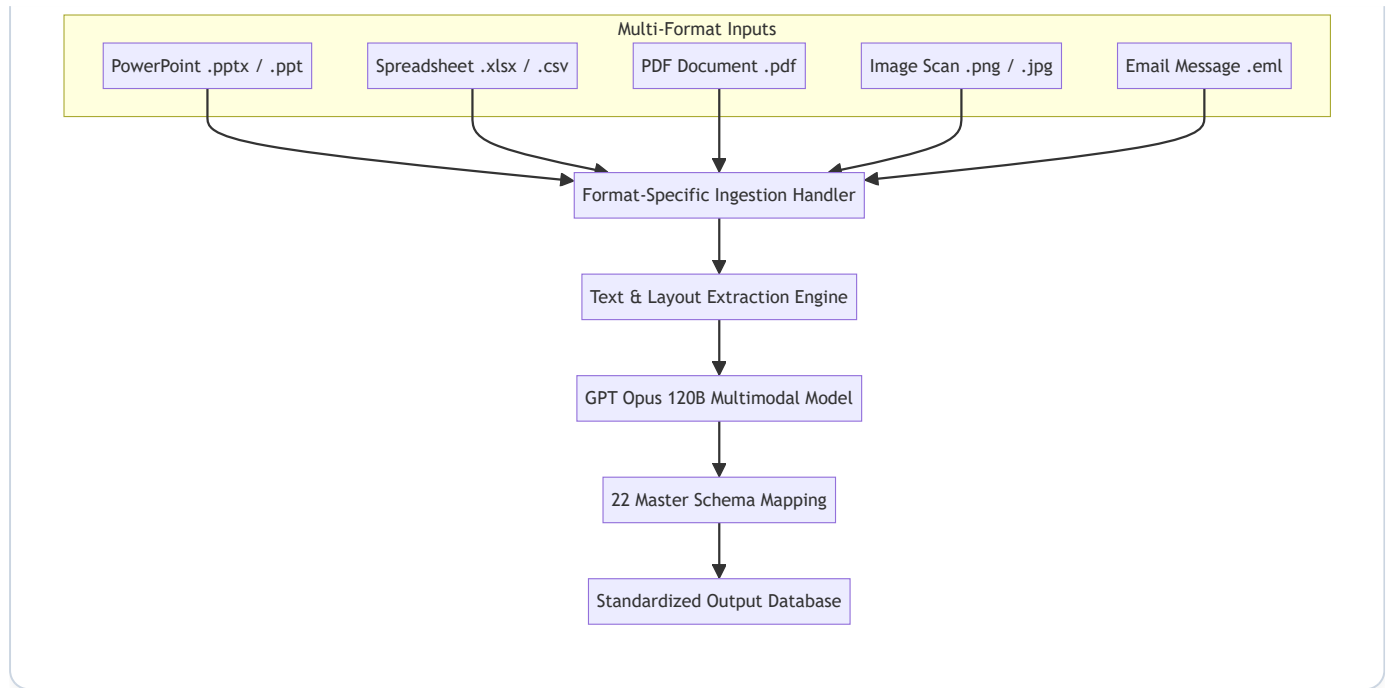
1.1 Objective

The Multi-Format File Processing System provides a unified ingestion framework capable of extracting operational maintenance records from multi-modal source documents.

1.2 Individual Module Flow Diagram



SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Ingestion Specifications by File Format

2.1 PowerPoint Presentation Decks (PPT / PPTX)

- **Requirements:**
- Extract text contents from slides, text boxes, callout shapes, and embedded tables.
- Parse slide titles to contextualize maintenance activity locations.

2.2 Spreadsheets (XLSX / XLS / CSV)

- **Requirements:**
- Auto-detect table boundaries, multi-header rows, and merged cells.
- Parse multiple worksheets within a single workbook.

2.3 PDF Inspection Reports (PDF)

- **Requirements:**
- Process both native digital PDFs and scanned image-based PDFs.
- Retain tabular layout structures during text extraction.

2.4 Diagnostic Images & Photos (PNG / JPG)

- **Requirements:**
- Extract text embedded within maintenance screenshots, equipment nameplates, and meter readings.
- Feed visual context directly to the multimodal AI processing pipeline.

2.5 Email Communications (EML)

- **Requirements:**
 - Extract email headers (Sender, Date, Subject) and body text.
 - Process attached documents (e.g., attached spreadsheets or PDF reports).
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3. QC Testing & Acceptance Criteria

- **Format Resilience Test:** The system MUST process all 5 supported file types in a single execution queue without failing.
- **Data Completeness Test:** Extracted fields from all file types MUST map into the 22 master schema attributes.
- **Error Handling:** Corrupt or unreadable files MUST generate a clear error message and isolate affected items into Quarantine.